

Granulocyte colony-stimulating factor and drugs elevating extracellular adenosine synergize to enhance haematopoietic reconstitution in irradiated mice

Pospíšil M, Hofer M, Znojil V, Netíková J, Vácha J, Holá J, Vacek A.
Granulocyte colony-stimulating factor and drugs elevating extracellular adenosine synergize to enhance haematopoietic reconstitution in irradiated mice.

Eur J Haematol 1998; 60: 172–180. © Munksgaard 1998.

Abstract: The activation of adenosine receptors has recently been demonstrated to stimulate haematopoiesis. In the present study, we investigated the ability of drugs elevating extracellular adenosine to influence curative effects of granulocyte colony-stimulating factor (G-CSF) in mice exposed to a sublethal dose of 4 Gy of ^{60}Co radiation. Elevation of extracellular adenosine in mice was induced by the combined administration of dipyridamole, a drug inhibiting the cellular uptake of adenosine, and adenosine monophosphate (AMP), an adenosine prodrug. The effects of dipyridamole plus AMP, and G-CSF, administered either alone or in combination, were evaluated. The drugs were injected to mice in a 4-d treatment regimen starting on d 3 after irradiation and the haematopoietic response was evaluated on d 7, 10, 14, 18 and 24 after irradiation. While the effects of G-CSF on the late maturation stages of blood cells, appearing shortly after the completion of the treatment, were not influenced by dipyridamole plus AMP, positive effects of the combination therapy occurred in the post-irradiation recovery phase which is dependent on the repopulation of haematopoietic stem cells. This was indicated by the significant elevation of counts of granulocyte-macrophage progenitor cells (GM-CFC) and granulocytic cells in the bone marrow (d 14), of GM-CFC (d 14), granulocytic and erythroid cells (d 14 and 18) in the spleen, and of neutrophils (d 18), monocytes (d 14 and 18) and platelets (d 18) in the peripheral blood. These effects suggest that the repopulation potential of the combination therapy lies in a common multilineage cell population. The results of this study implicate the promising possibility to enhance the curative effects of G-CSF under conditions of myelosuppressive states induced by radiation exposure.

**M. Pospíšil¹, M. Hofer¹, V. Znojil²,
J. Netíková¹, J. Vácha², J. Holá¹
and A. Vacek¹**

¹Institute of Biophysics, Academy of Sciences of the Czech Republic, Brno, Czech Republic and ²Medical Faculty, Masaryk University, Brno, Czech Republic

Key words: adenosine monophosphate – dipyridamole – G-CSF – myelosuppression – radiation – therapy

Correspondence: Dr M. Pospíšil, Institute of Biophysics, Academy of Sciences of the Czech Republic, Královopolská 135, 612 65 Brno, Czech Republic
Fax: +420-5-41211293

Accepted for publication 27 October 1997

Proliferation, differentiation and functionality of haematopoietic cells is under the control of a complex of regulatory factors, including the family of haematopoietic cytokines (1–3) and other regulators, such as eicosanoids, lactoferrin, etc. (4). These factors act in cooperative fashion, and evidence is accumulating on the possibility of enhancing the haematopoietic response by multiple growth factor combinations (1–3, 5–9). Previously we postulated a mechanism, neglected up to that

time, which could participate in the positive control of haematopoiesis *in vivo*, namely of the adenosine receptor signalling. We have shown that granulocyte colony-stimulating factor (G-CSF) and drugs elevating extracellular adenosine synergize to enhance granulopoiesis in normal mice (10). Experiments revealing these effects were based on the concept which views extracellular adenosine as an important regulatory factor acting through cell surface receptors coupled to a variety of effector

systems such as adenylate and guanylate cyclase, K^+ and Ca^{2+} channels, etc. (11–14). It has been demonstrated by *in vitro* experiments that purine nucleotides and nucleosides, including adenosine, are capable of increasing proliferation and differentiation of a wide range of cell types (15). The action of extracellular adenosine was proposed to represent a universal and primitive regulatory system modulating the cell growth (16). Data suggesting the expression of adenosine receptors on myeloid cell lineages (17) were the reason of using adenosine receptor signalling in combination with G-CSF. Furthermore, it was found that addition of adenosine or adenosine monophosphate to murine long-term bone marrow cultures resulted in increased granulopoiesis (18).

The purpose of experiments reported here was to better understand aspects of the priming effects of adenosine receptor signalling on G-CSF action *in vivo*. The experimental procedure inducing elevation of extracellular adenosine in mice utilized the previously described (10) combined administration of dipyridamole, a drug inhibiting adenosine uptake by cells and keeping it in the extracellular space (19), and adenosine monophosphate (AMP), serving as an adenosine prodrug. AMP was used instead of adenosine because of its greater solubility in water, which allowed administration of small injection volumes, thus avoiding loading the mice with a large volume of fluid. It can be assumed that exogenously administered AMP is rapidly metabolized extracellularly to adenosine by cell surface ectonucleotidase activity (20). Thus, dipyridamole and AMP act jointly in enhancing extracellular adenosine level. As demonstrated earlier (10), the combination of dipyridamole and AMP administered with G-CSF induces the most expressive stimulatory effects on haematopoiesis if compared with action of dipyridamole or AMP given alone.

The evidence of interactions between G-CSF and drugs elevating extracellular adenosine is not only of theoretical interest (cooperation of regulatory signals), but seems to be important for practical reasons. Clinical experience proved that G-CSF can augment haematopoiesis under a number of different conditions of primary and secondary bone marrow dysfunction (21). Dipyridamole and AMP are clinically available drugs and their capability to enhance G-CSF effects could not only increase the therapeutic potential of this growth factor, but also reduce the costs of G-CSF therapy. Because of the promising possibility of utilizing the combination of these drugs in the haematopoietic curative strategies, it appears important to explore their effects under conditions of myelosuppressive states. In the present study

we therefore investigated the ability of the combination of dipyridamole and AMP to influence the G-CSF therapy of radiation-induced haematopoietic injury in mice.

Material and methods

Mice

Male (CBA×C57BL/10) F_1 mice 3 months of age and weighing an average of 25 g were used. The mice were produced in a conventional manner by the institute facility. Standardized pelleted diet and HCl-treated tap water (pH 2–3) were given *ad libitum*. The mice were kept under controlled conditions (light:dark 12:12) and the temperature was maintained at $22\pm1^\circ\text{C}$. Animal experiments were conducted according to the principles given in the "Guide for the Care and Use of Laboratory Animals" issued by the Czech Society for Laboratory Animal Science.

Irradiation

Mice were given single total-body doses of 4 Gy from a ^{60}Co gamma-ray source at a dose rate of 0.28 Gy/min. During irradiation the mice were placed in ventilated Plexiglas containers. The dose of 4 Gy was found to be sublethal; no deaths of mice were recorded within 30 d after irradiation.

Drugs

Recombinant human G-CSF (Neupogen; purchased from F. Hoffman-LaRoche Ltd, Basel, Switzerland) was diluted with 5% glucose and injected subcutaneously at a single daily dose of 3 μg per mouse in a volume of 0.1 ml. Dipyridamole (Sigma, St Louis, MO, USA) was dissolved in 0.4% tartaric acid and injected subcutaneously at a single daily dose of 2 mg per mouse in a volume of 0.4 ml. Adenosine 5'-monophosphate sodium salt from yeast (Sigma) was dissolved in distilled water and injected intraperitoneally at a single daily dose of 5 mg free acid per mouse in a volume of 0.2 ml. Glucose (5%), tartaric acid (0.4%) and saline were used for control injections. In the combination treatment, dipyridamole was administered 20 min before AMP; G-CSF was administered 30 min after AMP. The doses of drugs and their administration schedule were based on previous experimental experience (10).

Haematological methods

Blood samples were taken from the tail vein. After blood sampling, the mice were killed by cervical

dislocation, femora and spleens were dissected and marrow and spleen cells were harvested by standard procedures. Blood cell counts and numbers of nucleated cells of the femoral marrow and the spleen were determined using a Coulter Counter (model ZF; Coulter Electronics, UK). Platelet counts were done by the standard direct microscopic method. Differential counts were performed on smear preparations stained with the May-Grünwald-Giemsa method. The differential counts of marrow and spleen cells included the morphologically recognizable granulocytic (myeloblasts through segmented neutrophils) and erythroid (proerythroblasts through orthochromatic erythroblasts) precursor cells. Bone marrow and spleen haematopoietic progenitor cells committed to granulocyte-macrophage development (granulocyte-macrophage colony-forming cells [GM-CFC]) were assayed using a semisolid plasma-clot technique. Briefly, femoral marrow and spleen cell suspensions were plated in quadruplicates using a 10% mouse lung-conditioned medium as a source of colony-stimulating factor. Colonies (>50 cells) were counted after 7 d of incubation in 37°C humidified environment containing 5% CO₂.

Study design

Drugs were administered in 4 daily injections from d 3 to d 6 after irradiation, always within 9–10 a.m. Mice received either the vehicle, dipyridamole plus AMP, G-CSF alone, or the combination of all drugs (combination therapy). Haematopoietic effects of the drug treatment of mice were determined at various time intervals after irradiation. No repeated sampling of blood from individual mice was used. Baseline values were collected from normal non-irradiated mice. Each individual experiment comparing the effects of drugs and their combinations included the parallel investigation of 3 drug-treated groups and 1 vehicle-treated control group. Experiments were repeated 2–4 times and all individual values were pooled. Five–20 mice per group and every time interval were used.

Statistics

The results are represented as the means±SEM of pooled data. The homogeneity of data from repeated experiments was verified. The results were subjected to analysis of variance followed by multiple comparisons using Tukey's test (CSS: Statistica, v 3.1, StatSoft, Inc, Tulsa, USA, 1992). The significance level was set at $p<0.05$; for the sake of simplicity only this value was used in all comparisons.

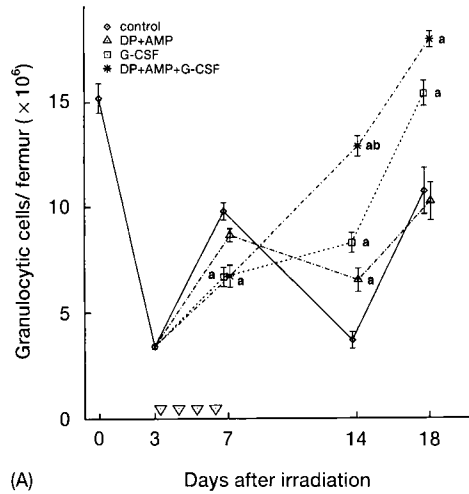
Results

Characteristics of the haematopoietic response to irradiation

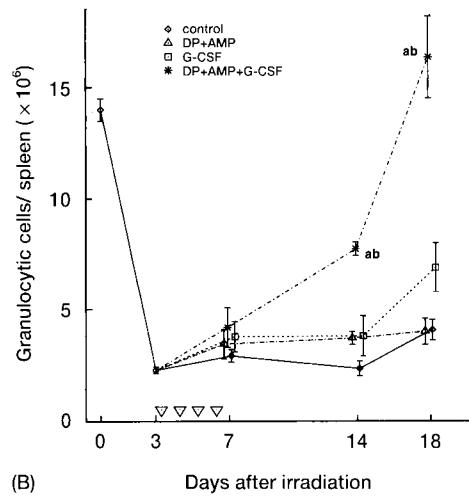
Data presented in Figs 1–3 and Table 1 indicate that irradiation of vehicle-treated control mice with a dose of 4 Gy induces the known quantitative changes of haematopoietic indices, including the initial drop followed by the more or less expressed rise which reflects the operation of regenerative processes. The degree and duration of the cell depletion was dependent on the cell type. The counts of neutrophils, erythrocytes, monocytes and platelets in peripheral blood recovered to normal values on d 24 after irradiation. Supranormal numbers of blood monocytes were produced during the recovery phase. In accordance with the radiobiological experience (22), the response of the myelocytic cell system to irradiation could be divided into several phases, i.e. first depletion phase, abortive rise (d 7–10), second depletion phase, and recovery phase (d 14–24).

Effects of the drugs on granulopoiesis

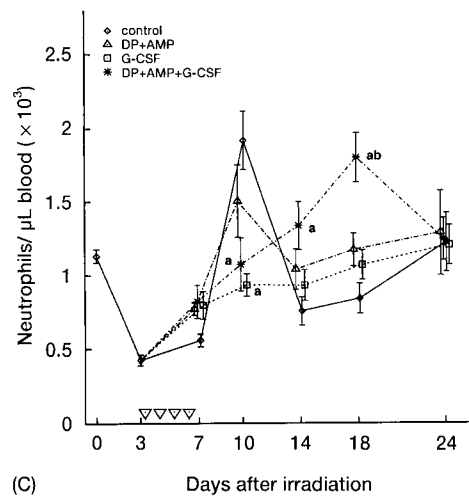
The effects of the drugs on bone marrow and spleen granulocytic cells and on neutrophil counts in peripheral blood of irradiated mice are illustrated in Fig. 1. On d 7 after the irradiation, i.e. 24 h after the completion of the drug therapy, G-CSF alone or the combination of all drugs significantly reduced the counts of granulocytic cells in the femoral marrow when compared with controls (Fig. 1A). No differences were found between the effects of both therapies. On d 14 and 18, however, the most effective recovery of granulocytic cells in the femoral marrow was observed in mice subjected to the combination therapy. On d 14, the combination therapy not only eliminated the second depletion phase, but significantly accelerated the recovery of granulocytic cells to an extent exceeding that seen with G-CSF alone. In the spleen, only the combination therapy significantly increased the counts of granulocytic cells on d 14 and 18 after irradiation (Fig. 1B). Treatment with G-CSF alone or with the combination of all drugs abrogated the abortive rise of neutrophils in blood, which was observed in controls on d 10 after irradiation (Fig. 1C); there were no significant differences between the effects of G-CSF alone and the effects of combination therapy. The lack of the abortive rise of blood neutrophils reflected the behaviour of granulocytic bone marrow cells, the number of which was reduced on d 7 in groups treated with G-CSF (see Fig. 1A). On d 14 and 18 after irradiation the combination therapy increased blood neutrophils most effectively, however. In mice treated with the combination of



(A) Days after irradiation

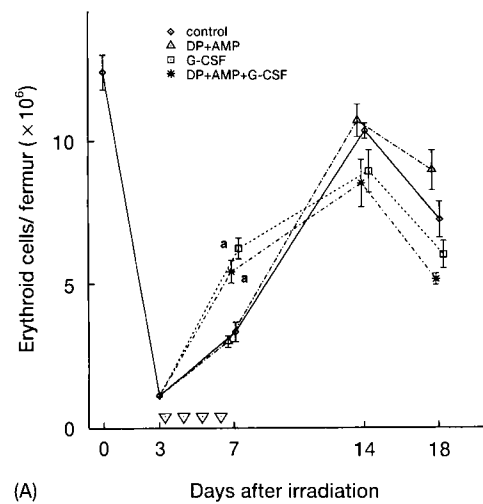


(B) Days after irradiation

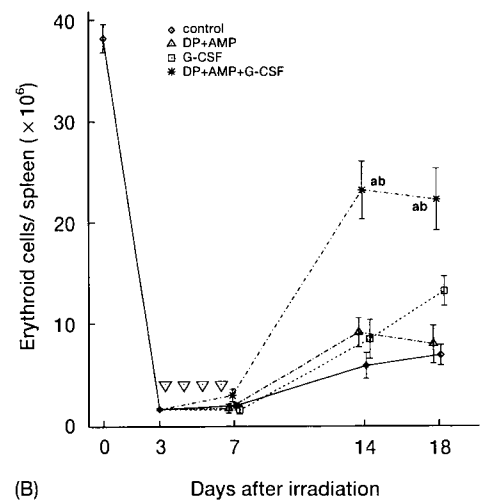


(C) Days after irradiation

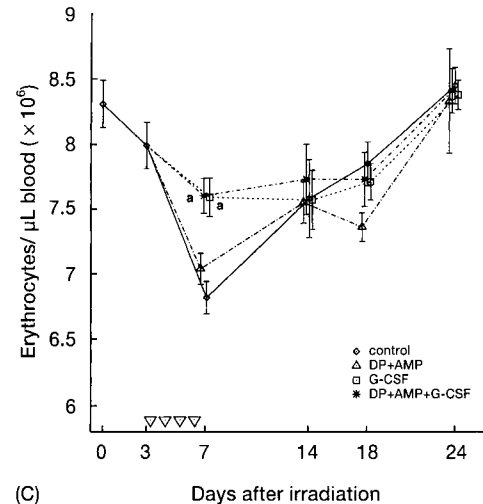
Fig. 1. Effects of the drug treatment on the post-irradiation response of granulocytic cells in femoral marrow (A) and spleen (B), and of peripheral neutrophils (C). Data are given as means $\pm$ SEM; 5–10 mice per point were used in case of tissue cellularity, 10–20 mice per point in case of blood cells. Baseline values are given on d 0. ∇ indicates time of drug treatment. Statistical significance ($p < 0.05$): a, compared with irradiated controls; b, compared with mice given G-CSF alone. DP, dipyridamole; AMP, adenosine monophosphate.



(A) Days after irradiation



(B) Days after irradiation



(C) Days after irradiation

Fig. 2. Effects of the drug treatment on the post-irradiation response of erythroid cells in the femoral marrow (A) and spleen (B), and of peripheral erythrocytes (C). Data are given as means $\pm$ SEM; 5–10 mice per point were used. Baseline values are given on d 0. ∇ indicates time of drug treatment. Statistical significance ($p < 0.05$): a, compared with irradiated controls; b, compared with mice given G-CSF alone. DP, dipyridamole; AMP, adenosine monophosphate.

all drugs, an overshoot of baseline was observed on d 18 after irradiation.

Effects of the drugs on erythropoiesis

Fig. 2 illustrates effects of drugs on the indices of erythropoiesis. In comparison with controls, G-CSF alone or the combination therapy significantly

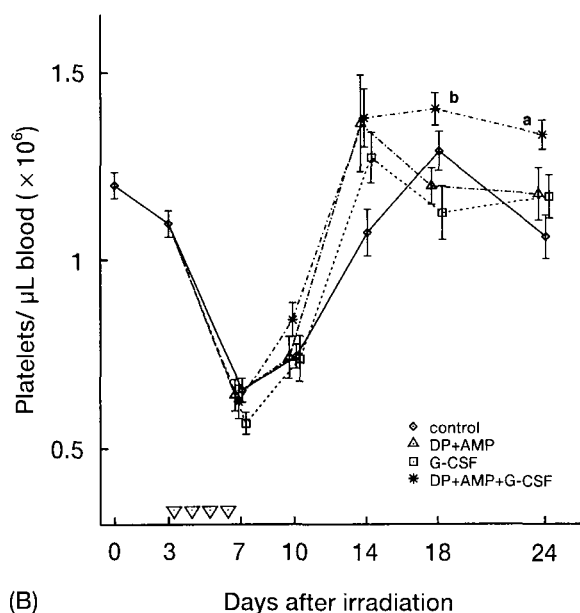
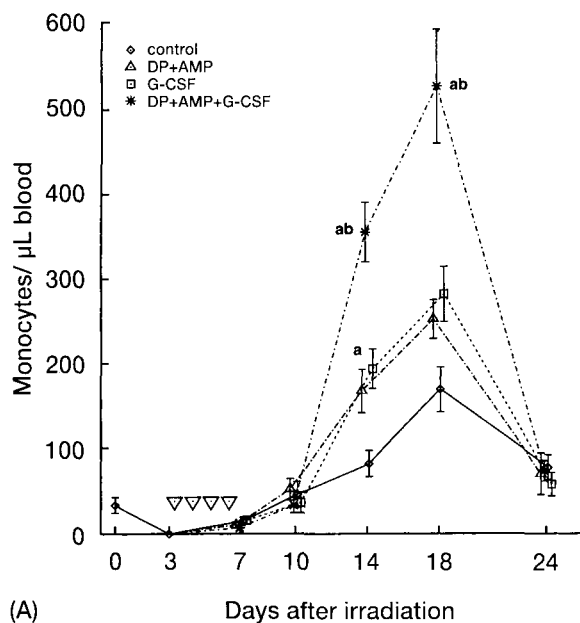


Fig. 3. Effects of the drug treatment on the post-irradiation response of peripheral monocytes (A) and platelets (B). Data are given as means $\pm$ SEM; 10–20 mice per point were used in case of monocytes, 8 mice per point in case of platelets. Baseline values are given on d 0. ∇ indicates time of drug treatment. Statistical significance ($p < 0.05$): a, compared with irradiated controls; b, compared with mice given G-CSF alone. DP, dipyridamole; AMP, adenosine monophosphate.

increased counts of erythroid cells in the femoral marrow on d 7 after irradiation (Fig. 2A). There were no differences between the effects of both therapies. At later intervals after irradiation, no significant effects of the treatment were detected. In the spleen, only the combination therapy significantly enhanced recovery of erythroid cells on d 14 and 18 compared to controls and animals treated with G-CSF alone (Fig. 2B). Erythrocytes in blood of control mice exhibited the lowest counts on d 7 after irradiation (Fig. 2C). This decline was significantly less pronounced in mice treated with G-CSF alone or with the combination of all drugs, thus reflecting an enhanced recovery of the bone marrow erythroid cells (see Fig. 2A). At later intervals after irradiation, no significant differences in blood erythrocyte counts between various groups of mice were found.

Effects of the drugs on blood monocytes, platelets and lymphocytes

As shown in Fig. 3A, monocytes disappeared from blood on d 3 after irradiation. Supranormal levels of monocyte counts occurred on d 14 and 18. G-CSF alone significantly enhanced the production of monocytes on d 14 compared to controls. The combination therapy was most effective, as shown by a significant increase of monocyte counts on d 14 and 18 compared to controls and animals treated with G-CSF alone.

Platelet counts were reduced to about 50% of baseline on d 7 after irradiation, without differences between the compared groups (Fig. 3B). In the recovery phase, no significant effect of G-CSF alone was found. The combination therapy appeared to increase the rebound thrombopoiesis, as indicated by the significantly higher platelet counts on d 18 compared to G-CSF alone, and on d 24 compared to controls.

Lymphocyte counts in blood of irradiated mice were not significantly influenced by the therapies used. On d 3 after irradiation, lymphocytes were reduced to about 16% of baseline. Up to d 24 after irradiation, these most sensitive blood cells recovered only to about 40% of the norm (data not shown).

Effects of the drugs on GM-CFC

Table 1 presents the effects of the drug treatment on the counts of GM-CFC in femoral marrow and spleen of irradiated mice. Dipyridamole plus AMP, as well as G-CSF given alone or in combination with dipyridamole plus AMP, significantly increased femoral GM-CFC counts on d 7 after irradiation; there were no differences between the effects of G-CSF alone and the effects of combination therapy.

Table 1. Effects of the drug treatment on GM-CFC counts in the femoral marrow and spleen of irradiated mice

	Control	DP + AMP	G-CSF	DP + AMP + G-CSF
GM-CFC/femur				
d 7 after irradiation	489 ± 28	1472 ± 48 ^a	2481 ± 161 ^a	2516 ± 155 ^a
d 14 after irradiation	11,163 ± 417	10,446 ± 419	12,324 ± 622	15,068 ± 557 ^{ab}
GM-CFC/spleen				
d 7 after irradiation	135 ± 10	144 ± 7	237 ± 21 ^a	172 ± 5 ^b
d 14 after irradiation	322 ± 18	646 ± 26 ^a	654 ± 31 ^a	876 ± 10 ^{ab}

Data are given as means ± SEM; 10–15 mice per group were used in case of femoral marrow, 5–10 mice in case of spleen. Statistical significance ($p < 0.05$): ^a, compared with irradiated controls; ^b, compared with mice given G-CSF alone. The mean counts of GM-CFC obtained in non-irradiated controls were 12,309 ± 385/femur; and 5421 ± 196/spleen. DP, dipyridamole; AMP, adenosine monophosphate.

As late as on d 14 after irradiation, the combination therapy was most effective, and significantly increased femoral GM-CFC counts compared to the action of G-CSF alone. In the spleen examined on d 7 after irradiation, only G-CSF alone significantly increased GM-CFC counts. On d 14, both the combination of dipyridamole plus AMP and G-CSF alone induced significant elevation of GM-CFC counts. However, the combination therapy was most effective and significantly increased the splenic GM-CFC counts compared to those obtained after administration of G-CSF alone.

Discussion

Damage to haematopoiesis induced in mice by a sublethal radiation dose of 4 Gy represents a suitable model which allows study of the main phases of the post-irradiation haematopoietic response and their modification by various treatment regimens. The treatment started on d 3 after exposure to allow some post-irradiation repair and regeneration of pluripotent and committed progenitor cells. A relatively short-term treatment regimen was used to investigate the rebound effects of the therapy. Under these conditions, only marginal stimulatory effects of dipyridamole plus AMP given without G-CSF were found. Therapy with G-CSF alone induced greater effects, but the best curative effects were obtained with the combination of all drugs. Kinetics of the post-irradiation response of the investigated cell lineages should be taken into consideration in discussion of the results.

The post-irradiation rise of granulocytic cells in bone marrow and peripheral blood of control mice, occurring on d 7 and 10, respectively, is a result of cell amplification in the compartment of precursor cells, and of the ensuing higher release of functional cells into blood. As summarized by Killmann (22),

the transitory nature of these effects probably relates to the insufficient supply of cells from the damaged and depleted stem cell pool. Cells that are responsible for initiating the abortive rise are rather severely damaged and capable of going only through a limited number of divisions. Thus, the second depletion of cells is a result of stem cell deficit. Provided that G-CSF induces an increase in the number of cell divisions and a shorter marrow transit time in the neutrophil precursor cell populations (23, 24), the lack of the abortive rise of granulocytic cells in bone marrow, and of neutrophils in blood of mice treated with G-CSF can be a consequence of rapid exhaustion of the morphologically recognizable cells due to the 4-d treatment regimen. Notably, the cell exhaustion did not include the bone marrow committed progenitor cells (GM-CFC), which were found to be increased by G-CSF on d 7 after irradiation. The phenomenon of the abrogation of the abortive rise of cell number, as well as that of the early enhancement of marrow GM-CFC counts by G-CSF, was not influenced by the administration of dipyridamole plus AMP. In the spleen, dipyridamole plus AMP reduced the stimulatory action of G-CSF on GM-CFC counts. The reason for this effect is not clear. Favourable effects of the combination therapy were observed in the post-irradiation recovery phase. On d 14 after irradiation, the counts of peripheral neutrophils and granulocytic cells in bone marrow and spleen of mice treated with the combination of drugs exceeded those observed after G-CSF alone and indicated accelerated initiation of the recovery. The stimulatory effects of this therapy persisted to d 18 after irradiation, with blood neutrophils exceeding baseline level. Also, GM-CFC counts in bone marrow and spleen, determined on d 14 post-irradiation, were higher after combination therapy compared with the action of G-CSF alone, although marrow GM-CFC had already reached baseline levels. In the interval of 14 d after irradiation, neutrophils in the blood increased 1.8-fold after the combination therapy compared to 1.2-fold increase after G-CSF alone, granulocytic cells in femoral marrow increased 3.5-fold after the combination therapy compared to 2.2-fold increase after G-CSF alone, granulocytic cells in spleen increased 3.3-fold after the combination therapy compared to 1.6-fold increase after G-CSF alone. Because the cell repopulation in later post-irradiation intervals reflects stem cell recovery, it is likely that the stimulatory effects of the drugs are mediated by their action on the pool of the more primitive haematopoietic cells which survive radiation exposure without serious damage and start to proliferate during the early post-irradiation period.

Although G-CSF is regarded generally as a neutrophil-specific regulator that potently stimulates the terminal stages of neutrophil development, its administration *in vivo* to normal and irradiated mice was found to stimulate not only granulopoiesis, but also proliferation of multipotential progenitor cells determined by the classic *in vivo* assay (splenic colony-forming cells) (25, 26). Furthermore, it has been shown that G-CSF enhances multipotential progenitor cell cycling in the marrow and spleen of normal mice (27). When using a novel *in vitro* stem cell assay, the cobblestone area forming cell assay, which enables quantification of all stem cell subsets, de Haan *et al.* (28) demonstrated that in normal mice the treatment with G-CSF induced a migration of stem cells from the marrow to the spleen, and an establishment of a higher, new steady state of the total stem cell compartment with the highest increase of the most committed stem cells. It is not clear whether the effects of G-CSF on multipotential progenitor cells *in vivo* are a result of its direct action or an indirect consequence of the increased production of granulocytes. However, it seems reasonable to hypothesize that the stimulatory interactions of dipyridamole plus AMP with G-CSF are realized at the level of the stem cell compartment. Support for this hypothesis is provided by our earlier observations that the protective treatment with dipyridamole plus AMP increases the formation of endogenous haematopoietic spleen colonies in gamma-irradiated mice (29, 30).

Results of our experiments indicate a preventive effect of G-CSF treatment on the reduction of erythroid cells in the bone marrow, and of erythrocyte counts in blood on d 7 after irradiation. Amplification of erythroid cells in the compartment of precursor cells can account for this effect. Stimulatory action of G-CSF *in vivo* on bone marrow erythroid progenitor and precursor cells was observed also by other authors (24, 25, 27). Coadministration of dipyridamole plus AMP did not modify these early effects of G-CSF. However, synergistic stimulatory action of the combination of all drugs on erythroid cells was detected in the spleen on d 14 and 18 after irradiation. The most reasonable explanation for this delayed effect is an enhanced recovery of multipotential and/or erythroid progenitors present in the spleen or immigrating from the bone marrow (28, 31) and differentiated in the splenic environment into the erythroid lineage.

An evidence of the pleiotropic action of the combination of drugs is provided by data illustrating the post-irradiation response of monocytes and platelets in peripheral blood. The observed stimulatory effects of G-CSF on the production of monocytes are in agreement with literature data

(23, 32, 33). Similarly, several studies demonstrated the stimulatory action of G-CSF on platelet production in myelosuppressed mice (9, 25, 26). Our experiments did not show significant effects of G-CSF alone on platelets probably because of a lower radiation dose and a short-term treatment regimen as opposed to those used in the above-mentioned studies. However, monocytopoiesis and thrombopoiesis were enhanced by the combination therapy. Because these actions were observed in the recovery phase, it is likely that the effects of the combination therapy involve a stimulation of the more primitive multipotential cell populations. As opposed to the granulopoietic system, the effects of the combination therapy appeared to be restricted to rebound monocytopoiesis and thrombopoiesis, without significantly altering the initiation of recovery.

Evidence presented in this study is consistent with an emerging role of extracellular adenosine as a potential mediator of cellular proliferation (10, 15, 16, 18). The mechanisms of the priming action of extracellular adenosine on G-CSF effects remain to be clarified. As discussed earlier, the signalling pathway of cyclic nucleotides could participate in this mechanism (10). Generally, synergistic effects between extracellular factors appear to result from the generation of complementary intracellular signals that act in concert to elicit the complete array of metabolic processes required for a proliferative response (34). Additionally, or alternatively, extracellular adenosine could stimulate target cells indirectly by producing secondary cytokines. Activation of adenosine receptors was found to increase erythropoietin production (35). It was demonstrated that adenosine enhances the production of interleukin-6 (36, 37).

The considerations presented above are based on the concept which views the cycling activity of cells as the main mechanism of the control of haematopoiesis. However, haematopoietic cytokines, including G-CSF, were found to promote viability of cells and to suppress apoptosis; they thus contribute to the production of daughter cells without the necessity to increase cell cycle activity (38, 39). It has been shown in our laboratory that the early pre- or post-irradiation administration of dipyridamole plus AMP to mice decreases the destruction of cells, as indicated by the thymidine level in plasma and the amount of soluble polynucleotides in thymus determined 4 h after exposure to 1 Gy (40, 41). Gasmi *et al.* (42) reported that diadenosine polyphosphates and adenosine triphosphate, acting on cells via purino receptors, interact with granulocyte-macrophage colony-stimulating factor to delay human neutrophil apoptosis. It remains to be ascertained whether mechanisms

suppressing apoptosis contribute to the synergistic effects of drugs demonstrated in the experiments presented here.

In general, our results suggest that drugs elevating extracellular adenosine can enhance the action of G-CSF in myelosuppressed mouse. The stimulatory action of the drug combination was observed after the 4-d treatment regimen, i.e. under conditions in which G-CSF given alone, induced only modest effects. The positive effects of the combination therapy occurred in the post-irradiation recovery phase, which is dependent on the repopulation of haematopoietic stem cells. This implies that targets of the combination therapy may be early haematopoietic progenitors, and that such an action could account for the observed multilineage cell response. It is worth mentioning that pleiotropic effects of G-CSF can occur also in man. For example, Broxmeyer *et al.* (43) have found that treatment of patients with G-CSF enhances cycling rates of granulocyte-macrophage, erythroid and multipotential progenitor cells. As indicated by experiments *in vitro*, stem cell proliferation requires costimulation by the early- and late-acting cytokines (1–3, 5). It has been shown that the early-acting haematopoietic cytokines, like stem cell factor and interleukin-6, act synergistically with G-CSF to enhance haematopoietic recovery in myelosuppressed mice (6, 7). It is perhaps reasonable to speculate that extracellular adenosine also plays a role as an early-acting and priming agent that modulates the effects of G-CSF.

Acknowledgements

This work was supported by the Grant Agency of the Czech Republic (Grant No. 306/96/1721).

References

- OGAWA M. Differentiation and proliferation of hematopoietic stem cells. *Blood* 1993; 81: 2844–2853.
- METCALF D. Hematopoietic regulators: redundancy or subtlety? *Blood* 1993; 82: 3515–3523.
- LOWRY PA. Hematopoietic stem cell cytokine response. *J Cell Biochem* 1995; 58: 410–415.
- BRENNAN JK, LICHTMAN MA, DIPERSIO JF, ABBOUD CN. Chemical mediators of granulopoiesis: a review. *Exp Hematol* 1980; 8: 441–464.
- IKEBUCHI K, CLARK SC, IHLE JN, SOUZA LM, OGAWA M. Granulocyte colony-stimulating factor enhances interleukin 3-dependent proliferation of multipotential hemopoietic progenitors. *Proc Natl Acad Sci USA* 1988; 85: 3445–3449.
- MOLINEUX G, MIGDALSKA A, SZMITKOWSKI M, ZSEBO KM, DEXTER TM. Effects on hematopoiesis of recombinant stem cell factor (ligand for c-kit) administered *in vivo* to mice either alone or in combination with granulocyte colony-stimulating factor. *Blood* 1991; 78: 961–966.
- PATCHEN ML, FISCHER R, MACVITTIE TJ. Effects of combined administration of interleukin-6 and granulocyte colony-stimulating factor on recovery from radiation-induced hemopoietic aplasia. *Exp Hematol* 1993; 21: 338–344.
- HEYWORTH CM, DEXTER TM, NICHOLLS SE, WHETTON AD. Combinations of colony-stimulating factors promote enhanced proliferative potential in enriched granulocyte-macrophage colony-forming cells. *Exp Hematol* 1994; 22: 1089–1094.
- GROSSMANN A, LENOX J, DEISHER TA, *et al.* Synergistic effects of thrombopoietin and granulocyte colony-stimulating factor on neutrophil recovery in myelosuppressed mice. *Blood* 1996; 88: 3363–3370.
- POSPÍŠIL M, HOŠER M, ZNOJIL V, VÁCHA J, NETÍKOVÁ J, HOLÁ J. Synergistic effect of granulocyte colony-stimulating factor and drugs elevating extracellular adenosine on neutrophil production in mice. *Blood* 1995; 86: 3692–3697.
- DALY JW. Adenosine receptors: targets for future drugs. *J Med Chem* 1982; 25: 197–207.
- NEWBY AC. Adenosine and the concept of “retaliatory metabolites”. *Trends Biochem Sci* 1984; 9: 42–44.
- STILES GL. Adenosine receptors and beyond: molecular mechanisms of physiological regulation. *Clin Res* 1990; 38: 10–18.
- BURNSTOCK G. Physiological and pathological roles of purines: an update. *Drug Devel Res* 1993; 28: 195–206.
- RATHBONE MP, MIDDLEMISS PJ, GYSBERS JW, DEFORGE S, COSTELLO P, DEL MAESTRO RF. Purine nucleosides and nucleotides stimulate proliferation of a wide range of cell types. *In Vitro Cell Dev Biol* 1992; 28A: 529–536.
- RATHBONE MP, CHRISTIANSON L, DEFORGE S, *et al.* Extracellular purine nucleosides stimulate cell division and morphogenesis: pathological and physiological implications. *Med Hypotheses* 1992; 37: 232–240.
- CRONSTEIN BN, ROSENSTEIN ED, KRAMER SB, WEISSMANN G, HIRSCHHORN R. Adenosine: a physiological modulator of superoxide anion generation by human neutrophils. Adenosine acts via an A₂ receptor on human neutrophils. *J Immunol* 1985; 135: 1366–1371.
- GUALTIERI RJ, BERNE RM, McGRATH HE, HUSTER WJ, QUESENBERRY PJ. Effect of adenine nucleotides on granulopoiesis and lithium-induced granulocytosis in long-term bone marrow cultures. *Exp Hematol* 1986; 14: 689–695.
- PLAGEMANN PGW, WOHLHUETER RM, WOFFENDIN C. Nucleoside and nucleobase transport in animal cells. *Biochim Biophys Acta* 1988; 947: 405–443.
- GORDON EL, PEARSON JD, DICKINSON ES, MOREAU D, SLAKEY LL. The hydrolysis of extracellular adenine nucleotides by arterial smooth muscle cells. *J Biol Chem* 1989; 264: 18986–18992.
- HARMENBERG J, HÖGLUND M, HELLSTRÖM-LINDBERG E. G- and GM-CFC in oncology and oncological haematology. *Eur J Haematol* 1994; 52: 1–28.
- KILLMANN SA. Radiation effects on the myelocytic cell renewal system. In: *Manual on Radiation Haematology*, Technical Reports Series No 123, Vienna: International Atomic Energy Agency, 1971: 89–97.
- LORD BI, MOLINEUX G, POJDA Z, SOUZA LM, MERMED JJ, DEXTER TM. Myeloid cell kinetics in mice treated with recombinant interleukin-3, granulocyte colony-stimulating factor (CSF), or granulocyte-macrophage CSF *in vivo*. *Blood* 1991; 77: 2154–2159.
- UCHIDA T, YAMAGIWA A. Kinetics of rG-CSF-induced neutrophilia in mice. *Exp Hematol* 1992; 20: 152–155.
- TANIKAWA S, NAKAO I, TSUNEOKA K, NARA N. Effects of recombinant granulocyte colony-stimulating factor (rG-CSF) and recombinant granulocyte-macrophage colony-stimulating factor (rGM-CSF) on acute radiation hematopoietic injury in mice. *Exp Hematol* 1989; 17: 883–888.

26. PATCHEN ML, MACVITTIE TJ, SOLBERG BD, SOUZA LM. Therapeutic administration of recombinant human granulocyte colony-stimulating factor accelerates hemopoietic regeneration and enhances survival in a murine model of radiation-induced myelosuppression. *Int J Cell Cloning* 1990; 8: 107–122.
27. BROXMEYER HE, WILLIAMS DE, COOPER S, HANGOC G, RALPH P. Recombinant human granulocyte-colony stimulating factor and recombinant human macrophage-colony stimulating factor synergize *in vivo* to enhance proliferation of granulocyte-macrophage, erythroid, and multipotential progenitor cells in mice. *J Cell Biochem* 1988; 38: 127–136.
28. DE HAAN G, DONTJE B, ENGEL C, LOEFFLER M, NIJHOF W. The kinetics of murine hematopoietic stem cells *in vivo* in response to prolonged increased mature blood cell production induced by granulocyte colony-stimulating factor. *Blood* 1995; 86: 2986–2992.
29. POSPÍŠL M, HOFER M, NETÍKOVÁ J, *et al.* Elevation of extracellular adenosine induces radioprotective effects in mice. *Radiat Res* 1993; 134: 323–330.
30. POSPÍŠL M, HOFER M, ZNOJIL V, VÁCHA J, NETÍKOVÁ J, HOLÁ J. Radioprotection of mouse hemopoiesis by dipyrindamole and adenosine monophosphate in fractionated treatment. *Radiat Res* 1995; 142: 16–22.
31. POJDA Z, MOLINEUX G, DEXTER TM. Hemopoietic effects of short-term *in vivo* treatment of mice with various doses of rhG-CSF. *Exp Hematol* 1990; 18: 27–31.
32. TAMURA M, HATTORI K, NOMURA H, *et al.* Induction of neutrophilic granulocytosis in mice by administration of purified human native granulocyte colony-stimulating factor (G-CSF). *Biochem Biophys Res Commun* 1987; 142: 454–460.
33. GILMORE GL, DEPASQUALE DK, FISCHER BC, SHADDUCK RK. Enhancement of monocytopoiesis by granulocyte colony-stimulating factor: evidence for secondary cytokine effects *in vivo*. *Exp Hematol* 1995; 23: 1319–1323.
34. ROZENGURT E. Early signals in the mitogenic response. *Science* 1986; 234: 161–166.
35. NAGASHIMA K, KARASAWA A. Modulation of erythropoietin production by selective adenosine agonists and antagonists in normal and anemic rats. *Life Sci* 1996; 59: 761–771.
36. BOUMA MG, STAD RK, VAN DEN WILDENBERG FAJM, BUURMAN WA. Differential regulatory effects of adenosine on cytokine release by activated human monocytes. *J Immunol* 1994; 153: 4159–4167.
37. RITCHIE PK, SPANGELO BL, KRZYMOWSKI DK, ROSSITER TB, KURTH E, JUDD AM. Adenosine increases interleukin 6 release and decreases tumour necrosis factor release from rat adrenal zona glomerulosa cells, ovarian cells, anterior pituitary cells, and peritoneal macrophages. *Cytokine* 1997; 9: 187–198.
38. WILLIAMS GT, SMITH CA, SPOONER E, DEXTER TM, TAYLOR DR. Haemopoietic colony stimulating factors promote cell survival by suppressing apoptosis. *Nature* 1990; 343: 76–79.
39. DE HAAN G, DONTJE B, NIJHOF W. Concepts of hemopoietic cell amplification. Synergy, redundancy and pleiotropy of cytokines affecting the regulation of erythropoiesis. *Leuk Lymph* 1996; 22: 385–394.
40. HOŠEK B, BOHÁČEK J, ŠIKULOVÁ J, POSPÍŠL M, VACEK A. Protection of early cellular damage in 1 Gy-irradiated mice by the elevation of extracellular adenosine. *Radiat Environ Biophys* 1992; 31: 289–297.
41. BOHÁČEK J, HOŠEK B, POSPÍŠL M. Post-irradiation administration of adenosine monophosphate combined with dipyrindamole reduces early cellular damage in mice. *Life Sci* 1993; 53: 1317–1324.
42. GASMI L, MCLENNAN AG, EDWARDS SW. The diadenosine polyphosphates Ap₃A and Ap₄A and adenosine triphosphate interact with granulocyte-macrophage colony-stimulating factor to delay neutrophil apoptosis: implications for neutrophil : platelet interactions during inflammation. *Blood* 1996; 87: 3442–3449.
43. BROXMEYER HE, BENNINGER L, PATEL SR, BENJAMIN RS, VADHAN-RAJ S. Kinetic response of human marrow myeloid progenitor cells to *in vivo* treatment of patients with granulocyte colony-stimulating factor is different from the response to treatment with granulocyte-macrophage colony-stimulating factor. *Exp Hematol* 1994; 22: 100–102.